

✓ CLIENT REQUIREMENT SUMMARY – RFID Solution

Current Situation

The client currently operates a **manual paper-based gate pass system** for employees who need to take company laptops outside the premises.

Current Process:

- System generates a gate pass.
- Divisional management approves it.
- Gate pass is printed.
- Security verifies and stamps at exit.
- On return, same paper is stamped again.
- Status updated manually in system.

Challenges:

- Paper loss or damage
 - Manual stamping dependency
 - Delayed or incorrect status updates
 - Limited real-time visibility
 - Security difficulty in tracking asset movement
 - Risk of unauthorized movement
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🎯 CLIENT OBJECTIVE

To implement a **paperless RFID-enabled automated laptop movement control system** that:

- Assigns RFID tags to each laptop
- Detects movement automatically at gates
- Updates asset status in real time
- Logs all transactions centrally
- Generates alerts for unauthorized movements
- Provides centralized monitoring dashboard
- Manages employee-laptop mapping
- Supports employee lifecycle (Active/Inactive)

- Enables future HCM integration
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ANTIGRAVITY PROPOSED SOLUTION

MODULE 1: EMPLOYEE MANAGEMENT

Employee Master Management

- Bulk employee upload (Excel)
- Manual employee creation
- Fields:
 - Employee ID
 - Employee Name
 - Department
 - Division
 - Designation
 - Contact details
 - Status (Active / Inactive)
- Edit employee details
- Activate / Deactivate employee
- Employee history log

Asset Tagging During Employee Creation

- Assign laptop during employee creation
- Map RFID tag to laptop
- Map laptop to employee
- Edit mapping later
- Reassign asset option
- Multiple assets (if required – configurable)

Employee Status Logic

If employee is:

- Marked Inactive:

- Movement blocked
 - Alert triggered if movement attempted
 - Asset recovery workflow initiated
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MODULE 2: ASSET / LAPTOP MANAGEMENT

Laptop Master

- Asset ID
- Serial Number
- Brand / Model
- RFID Tag ID (unique)
- Purchase details
- Warranty info
- Status:
 - Inside
 - Outside
 - Overdue
 - Unauthorized
 - Inactive

Asset Controls

- Edit asset
 - Reassign asset
 - Deactivate asset
 - View complete movement history
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MODULE 3: RFID GATE AUTOMATION

RFID Tagging

- Each laptop tagged with unique RFID
- RFID mapped in system
- Tamper detection (if supported)

RFID Readers

Installed at:

- Entry gates
- Exit gates

System behavior:

- Auto-detect laptop movement
 - Detect direction (Entry/Exit)
 - Real-time system update
 - Instant validation against approval database
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MODULE 4: DIGITAL GATE PASS WORKFLOW

Exit Process

- Employee requests digital gate pass
- Approval workflow:
 - Divisional Manager
 - Optional Admin approval
- Valid From / Valid Till defined

If Approved:

- RFID exit allowed
- Movement logged as Authorized Exit

If Not Approved:

- Marked Unauthorized
- Alert triggered

Re-entry Process

- RFID detects laptop on entry
 - Status auto-updated to Inside
 - Gate pass auto-closed
 - No manual stamping required
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MODULE 5: CENTRALIZED MONITORING DASHBOARD

Real-Time Dashboard Widgets

- Laptops Inside Premises
 - Laptops Outside Premises
 - Overdue Returns
 - Unauthorized Attempts
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DASHBOARD DRILLDOWN DETAILS

1 Laptops Inside Premises

Logic:

- Last movement = Entry
- No open gate pass
- Status = Inside

Drilldown Details (When Clicked):

Shows:

- Employee ID
- Employee Name
- Department
- Division
- Laptop Asset ID
- Serial Number
- RFID Tag ID
- Last Gate Name
- Last Movement Date & Time
- Gate Pass Status (Closed / Not Required)
- Assigned Date
- Asset Current Status

Filters Available:

- Department
- Division
- Gate
- Date Range
- Employee Name
- Asset ID

Export Option:

- Excel / PDF
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2 Laptops Outside Premises

Logic:

- Last movement = Exit
- Valid gate pass exists
- Not yet re-entered
- Status = Outside

Drilldown Details:

Shows:

- Employee ID
- Employee Name
- Department
- Division
- Laptop Asset ID
- Serial Number
- RFID Tag ID
- Exit Gate Name
- Exit Date & Time
- Gate Pass Valid Till
- Remaining Valid Time
- Approved By

- Approval Date
- Contact Number
- Current Status

Filters:

- Valid Till Date
- Department
- Gate
- Employee
- Division

Export:

- Excel / PDF
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3 Overdue Returns

Logic:

- Last movement = Exit
- Current time > Gate Pass Valid Till
- No re-entry detected

Drilldown Details:

Shows:

- Employee ID
- Employee Name
- Department
- Division
- Laptop Asset ID
- Serial Number
- Exit Gate
- Exit Date & Time
- Gate Pass Expiry Time
- Overdue Duration (Hours / Days auto-calculated)

- Approved By
- Escalation Status
- Alert Triggered (Yes/No)

Highlighting Rules:

- 0–4 hours overdue → Warning
- 4–24 hours → Medium Risk
- 24+ hours → Critical

Filters:

- Overdue Duration
- Department
- Division
- Gate
- Approval Authority

Export:

- Excel / PDF
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4 Unauthorized Attempts

Logic:

Triggered when:

- Exit without approval
- Entry attempt without valid mapping
- Inactive employee movement
- RFID tag not registered
- Expired gate pass used

Drilldown Details:

Shows:

- Employee ID (if identified)
- Employee Name (if mapped)
- Laptop Asset ID

- RFID Tag ID
- Gate Location
- Attempt Type (Unauthorized Exit / Entry)
- Attempt Date & Time
- Reason (No Approval / Expired / Inactive Employee / Unknown Tag)
- Security Action Taken
- Alert Status
- CCTV Reference ID (if integrated)

Filters:

- Date Range
- Gate
- Attempt Type
- Department
- Status (Resolved / Pending)

Export:

- Excel / PDF
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MODULE 6: ALERT SYSTEM

Automated alerts triggered for:

- Unauthorized exit
- Unauthorized entry
- Expired gate pass
- Inactive employee movement
- Unmapped RFID
- Overdue return

Alert Channels:

- Email
- SMS (optional)
- System notification

- Optional alarm at gate
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MODULE 7: AUDIT & REPORTING

Standard Reports

- Daily movement report
- Employee-wise movement history
- Asset-wise movement history
- Gate-wise movement report
- Overdue report
- Unauthorized attempts report
- Approval history report
- Inactive employee attempt report

All reports:

- Filterable
- Exportable
- Fully auditable

Audit Logs Include:

- Who approved
 - When approved
 - Gate used
 - RFID detected
 - Status change timestamps
 - System action logs
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MODULE 8: ROLE-BASED ACCESS CONTROL

Roles:

- Super Admin
- Security Admin
- IT Asset Admin

- Divisional Manager
- HR
- Security Guard (View Only)

Permissions Configurable:

- Approve gate pass
 - Edit employee
 - Assign asset
 - View dashboard
 - View reports
 - Manage RFID mapping
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MODULE 9: PHASE 2 – HCM INTEGRATION

Integration Scope

- Auto-fetch employee data from HCM
- Auto-sync:
 - New employees
 - Status changes
 - Transfers
 - Resignations
- Auto-mark inactive employees
- Optional asset sync with HCM

Integration Methods

- API-based integration
- Scheduled sync
- Webhook-based real-time sync